

Final Project

STAT452 – Applied Statistics for Engineering and Scientists II

Academic Year 2025–2026

Overview

The Final Project is a group assignment consisting of two parts. **Part 1** focuses on the full supervised-learning workflow on a real tabular dataset: exploratory data analysis, data cleaning, model building with regularization, and evaluation. **Part 2** is an open extension in which each group deepens the analysis with more advanced methods and a critical discussion. Each group selects **one** dataset from the list in Section 2 (no two groups may take the same dataset; assignment is first-come-first-served).

1. Logistics

- **Group size:** 4–5 students. Register your group and chosen dataset via the course form.
- **Deliverables:** (1) a written report (PDF, max. 20 pages excluding appendix); (2) reproducible source code (Jupyter notebook or R Markdown); (3) a 10-minute presentation.
- **Reproducibility:** fix a random seed; the notebook must run top-to-bottom without manual intervention.
- **Academic integrity:** you may use standard libraries (scikit-learn, statsmodels, glmnet, ...), but all analysis, interpretation, and writing must be your own. Cite every external source.

2. Datasets for Part 1

Each dataset below is a small-to-medium tabular dataset from the UCI Machine Learning Repository, chosen so that the full Part 1 workflow (visualization, outlier / missing-value handling, feature selection, and regularization) can be carried out meaningfully. Both regression and classification problems are represented.

2.1 Project 01 — Concrete Compressive Strength

Task: Regression

Size: 1030×8

Target: Concrete compressive strength (MPa), continuous

Predict the compressive strength of concrete from its mix composition (cement, blast-furnace slag, fly ash, water, superplasticizer, coarse / fine aggregate in kg/m^3 , and age in days). The relationship between age, ingredients and strength is strongly nonlinear, making the dataset ideal for practising transformations and regularization.

Source: <https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength>

2.2 Project 02 — Airfoil Self-Noise

Task: Regression

Size: 1503×6

Target: Scaled sound pressure level (dB), continuous

Predict the noise level generated by NACA airfoil blade sections tested in an anechoic wind tunnel, from frequency, angle of attack, chord length, free-stream velocity, and suction-side displacement thickness. Few features but nonlinear relationships — good for residual analysis and

feature transformation.

Source: <https://archive.ics.uci.edu/dataset/291/airfoil+self+noise>

2.3 Project 03 — Wine Quality (Red)

Task: Regression *or* Classification

Size: 1599×11

Target: Quality score (0–10), or good / poor class

Predict wine quality from physicochemical measurements (acidity, residual sugar, chlorides, sulphur dioxide, density, pH, sulphates, alcohol). The dataset can be treated as regression (predict the score) or classification (good vs. poor). Features are strongly correlated, so it is well suited to illustrating Ridge / Lasso / Elastic Net.

Source: <https://archive.ics.uci.edu/dataset/186/wine+quality>

2.4 Project 04 — Breast Cancer Wisconsin (Diagnostic)

Task: Classification (binary)

Size: 569×30

Target: Diagnosis: malignant / benign

Classify tumours as malignant or benign from 30 features computed from digitised images of fine-needle aspirates. A biomedical classification task where the choice and interpretation of metrics (precision, recall, F_1 , ROC-AUC) matters because of class imbalance and asymmetric misclassification costs.

Source: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

2.5 Project 05 — Energy Efficiency

Task: Regression (two targets)

Size: 768×8

Target: Heating Load and Cooling Load, continuous

Predict the heating and cooling loads of buildings from eight geometric / design features (relative compactness, surface area, wall / roof area, height, orientation, glazing area and distribution). Two continuous targets allow comparison of two regression models; several ordinal / categorical features give practice in encoding.

Source: <https://archive.ics.uci.edu/dataset/242/energy+efficiency>

Dataset summary

#	Dataset	Task	Size
01	Concrete Compressive Strength	Regression	1030×8
02	Airfoil Self-Noise	Regression	1503×6
03	Wine Quality (Red)	Regression / Classification	1599×11
04	Breast Cancer Wisconsin	Classification	569×30
05	Energy Efficiency	Regression (2 targets)	768×8

3. Part 1 — Required Workflow

Carry out and document the following steps on your chosen dataset.

1. **Exploratory Data Analysis.** Describe the variables and their types. Produce informative visualizations (distributions, pairwise relationships, correlation structure). Summarise what the plots reveal about the modelling problem.
2. **Data cleaning.** Detect and handle missing values and outliers, justifying each decision. Apply any transformations (scaling, log, encoding of categorical variables) that the data motivate, and explain why.
3. **Feature selection.** Investigate which predictors matter, using at least one principled approach (e.g. correlation / VIF screening, stepwise selection, or embedded selection via Lasso). Report and interpret the result.
4. **Modelling with regularization.** Fit a baseline model and at least two regularized models (Ridge, Lasso, and/or Elastic Net for regression; regularized logistic regression for classification). Tune the regularization strength by cross-validation and report how the coefficients change.
5. **Evaluation.** Choose evaluation metrics appropriate to the task (RMSE / MAE / R^2 for regression; accuracy, precision, recall, F_1 , ROC-AUC for classification) and justify the choice. Compare models fairly on a held-out test set and discuss the bias–variance trade-off you observe.

4. Part 2 — Extension

Students find their own dataset (encouraged to use real-world data) appropriate for experimental design analysis. The goal is to perform a comprehensive study using two-factor ANOVA or 2^k factorial design (where $k \geq 2$) to test hypotheses, analyze variance, and interpret results.

Requirements:

- **Dataset selection:**

- Find a suitable dataset (preferably from public sources like Kaggle, government databases, or research papers). The dataset should have: at least two categorical factors (for ANOVA) or k factors with 2 levels each (for 2^k factorial design) and a continuous response variable for analysis.
- Justify why the dataset is appropriate for the chosen experimental design.

- **Research Questions & Hypotheses:**

- Define clear research questions related to the factors and response variable.
- Formulate null and alternative hypotheses for the experiment.

- **Descriptive Statistics.**

- Summary statistics (mean, median, variance, etc.).
- Visualizations (box plots, interaction plots, histograms, etc.).
- Check assumptions (normality, homogeneity of variance, independence).

- **ANOVA / Factorial Design Analysis**

- Conduct ANOVA analysis.

- Interpret the main effects and interaction effects (if applicable).
- Check model adequacy (residual analysis, diagnostic plots).
- **Post-hoc Analysis:**
 - If ANOVA results are significant, perform post-hoc tests (Tukey’s HSD, Bonferroni, etc.).
 - Discuss practical significance.
- **Conclusion & Evaluation**
 - Summarize key findings.
 - Discuss limitations and potential improvements.
 - Provide actionable insights based on the results.

5. Expectations & Grading

Expectations

A strong project is judged less by raw predictive accuracy than by the **quality of reasoning**: are the modelling choices justified, are the metrics appropriate to the problem, and is the discussion honest about limitations? Clear writing and fully reproducible code are essential.

Component	Weight
Part 1 — EDA, cleaning, feature selection	25%
Part 1 — Modelling with regularization & evaluation	25%
Part 2 — Extension (depth and correctness)	25%
Report clarity, structure, and interpretation	15%
Reproducibility of code and presentation	10%